

SUMMER INTERNSHIP PROJECT REPORT

Fake News Detection and Evaluation with Confusion Matrix

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1. Abstract

As correctly said by Abraham Lincoln, “Don’t believe everything you read on the Internet, just because there’s a picture with a quote next to it!” The rapid spread of misinformation through digital media has made “fake news detection” an important application of Machine Learning and Natural Language Processing, in today’s world. The Project: **“Fake News Detection and Evaluation with Confusion Matrix”** aims to classify news articles as either fake or genuine using a supervised learning approach. The dataset was prepared by combining thousands of separate collections of fake and true news articles assigning appropriate class labels. After merging and cleaning the data, text preprocessing techniques such as lowercasing, URL removal, and the elimination of special characters were applied to improve data quality. The cleaned text data was then converted into numerical representations using the TF-IDF (Term Frequency – Inverse Document Frequency) vectorization technique. The dataset was further divided into training and testing sets to evaluate the model performance. Two Machine Learning (ML) algorithms, Logistic Regression and Decision Tree Classifier, were trained and tested on the processed data. Model performance was assessed using classification metrics and accuracy scores. Finally, Hyperparameter tuning was performed using GridSearchCV to optimize the Logistic Regression model. The results demonstrated that ML techniques can effectively distinguish between fake and real news articles with extremely high accuracy!

2. Introduction

The rapid growth of digital media has made information more accessible than ever! However, it has also led to the widespread circulation of fake news, which can mislead people and influence public opinion. Detecting fake news thus has all the more become an important application of Machine Learning and Natural Language Processing.

The objective of this project is to develop a Machine Learning model that can classify news articles as either fake or genuine based on their textual content. The project involved data preparation, text preprocessing, feature extraction using TF-IDF, and the implementation of classification algorithms namely, Logistic Regression and Decision Tree. Finally, the performance of the models was evaluated using accuracy and classification metrics; while GridSearchCV was used to optimize the Logistic Regression model.

This project provided practical exposure to the complete ML workflow, including data cleaning, feature engineering, model training, evaluation, and hyperparameter tuning. The primary purpose of doing this project was to gain hands-on experience in applying ML techniques to solve a real-world problem. With the growing influence of digital media, identifying fake news has become increasingly pivotal for ensuring the reliability of information. Through this project, I explored the complete ML pipeline. Beyond building an effective fake news classifier, the project helped me to strengthen my understanding of how ML can be used to analyze large volumes of textual data and support informed decision-making.

Honestly, “without big data, you are blind and deaf and in the middle of a freeway.” And, “Machine Intelligence is the last invention that humanity will ever need to make!”

The first two weeks of the Summer Internship Training turned out to be extremely helpful as, the concepts learned during these sessions provided the foundation of Machine Learning which was an essential for successfully completing the Fake News Detection project. The following topics were covered:

- Python Basics (Data Types, Variables, Lists, and Loops)
- Python Data Structures
- Functions, Classes, and Object-Oriented Programming (OOPS)
- NumPy and Pandas
- Machine Learning Overview
- Machine Learning Fundamentals
- Large Language Model (LLM) Fundamentals
- Communication Skills

3. Project Objective

Project Goal: AUTOMATED FAKE NEWS CLASSIFICATION

The primary goal of this project is to design and evaluate an ML-based system capable of automatically distinguishing fake news articles from genuine news articles using Natural Language Processing techniques.

OBJECTIVES:

- i. To build a supervised Machine Learning model using a labeled dataset of news articles.
- ii. To preprocess and clean textual data by applying techniques such as lowercasing, URL removal, and elimination of unnecessary characters in order to improve data quality.
- iii. To convert unstructured news content into numerical representations using the TF-IDF (Term Frequency- Inverse Document Frequency) vectorization technique.
- iv. To train and evaluate classification algorithms, namely Logistic Regression and Decision Tree Classifier, for the task of fake news detection.
- v. To assess model performance using standard evaluation metrics such as accuracy, precision, recall, and F1-score, including a confusion matrix to analyze true vs. false positives and negatives; and to improve model performance through hyperparameter-tuning using GridSearchCV.

4. Methodology

4.1 Data Collection

The project utilized a publicly available dataset consisting of news articles categorized as either fake or genuine. The dataset was provided in two separate files: **Fake.csv**, containing fake news articles, and **True.csv**, containing genuine news articles. Each record included information such as the news title, article text, subject category, and publication date. Since the dataset was already labeled, no primary data collection or survey was required. The analysis thus, was based entirely on **secondary data** obtained from existing sources.

4.2 Data Preparation and Labeling

To prepare the dataset for ML analysis, both the datasets were loaded into Python using Pandas library. A new column named **class** was created to represent the target variable.

- Fake news articles were assigned the label **1**
- Genuine (True) news articles were assigned the label **0**

The two datasets were then merged into a single DataFrame and randomly shuffled to ensure that the observations were distributed uniformly throughout the dataset.

4.3 Data Cleaning

Several preprocessing steps were performed to improve the quality of the dataset.

- Unnecessary columns such as **title**, **subject**, and **date** were removed from the dataset.
- Rows containing missing values in the text field were identified and removed.
- The dataset index was reset after cleaning to maintain consistency.

The resulting dataset contained only two variables: **text** (news article content) and **class** (target variable).

4.4 Text Preprocessing

Since machine learning algorithms cannot directly process raw text, the textual data was cleaned and standardized using a custom preprocessing function.

The following operations were performed:

- Conversion of all text to lowercase
- Removal of URLs and web links
- Removal of special characters and punctuation marks
- Replacement of multiple spaces with a single space
- Removal of leading and trailing spaces

These steps helped reduce noise in the dataset and improved the quality of feature extraction.

4.5 Training and Testing Data Split

The cleaned dataset was then divided into training and testing subsets using the `train_test_split()` function from Scikit-learn.

- ✓ **Training Set:** 75% of the data
- ✓ **Testing Set:** 25% of the data

The training data was used to build the machine learning models, while the testing data was used to evaluate their performance on unseen observations.

4.6 Feature Extraction using TF-IDF

The textual data was converted into numerical features using **Term Frequency–Inverse Document Frequency (TF-IDF)** vectorization. TF-IDF measures the importance of a word in a document relative to its frequency across all documents. Words that occur frequently within a specific article but less frequently across the entire dataset receive higher weights.

The TF-IDF vectorizer was fitted on the training data and subsequently used to transform both training and testing datasets into numerical feature matrices.

4.7 Machine Learning Models

Two supervised machine learning models were developed and evaluated.

4.7.1 Logistic Regression

Logistic Regression was used as the primary classification model due to its effectiveness in binary classification problems. The model was trained using the TF-IDF feature matrix and then used to predict whether a news article was fake or true!

Model performance was evaluated using:

- ✓ Precision
- ✓ Recall
- ✓ F1-score
- ✓ Accuracy

4.7.2 Decision Tree Classifier

A Decision Tree Classifier was also trained on the same feature set. The model learns decision rules from the training data and classifies articles based on these learned patterns. The accuracy score obtained from the testing dataset was used to evaluate its performance.

4.8 Hyperparameter Tuning

To improve model performance, **GridSearchCV** was applied to the Logistic Regression model. The following parameter combinations were explored:

- ✓ $C = \{0.1, 1, 10\}$
- ✓ Solver = {liblinear, lbfgs}

Five-fold cross-validation was used to identify the optimal combination of parameters. The best-performing configuration obtained was: **C = 10, Solver = liblinear**.

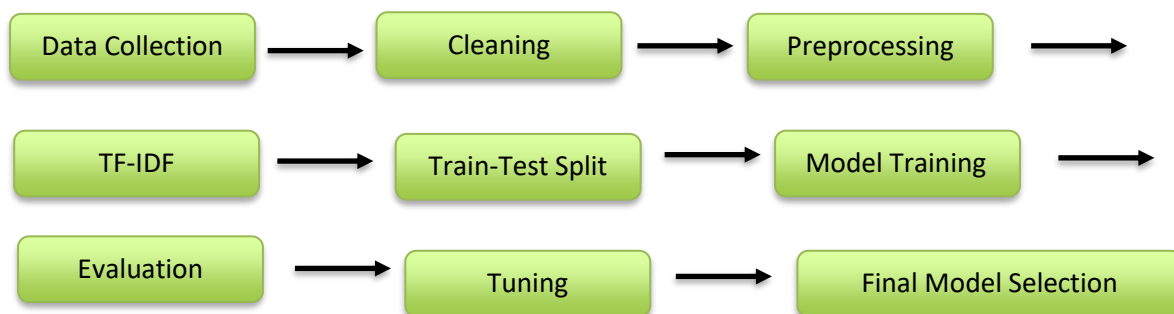
This optimized model demonstrated excellent classification performance on the testing dataset.

4.9 Software and Tools used

The project was implemented using the following tools and libraries:

Python, Google Colab, Pandas, NumPy, Scikit-learn, Regular Expressions(re), TF-IDF Vectorizer, Logistic Regression, Decision Tree Classifier and GridSearchCV.

4.10 Workflow of the Project



4.11 Model Validation

Model validation was carried out using a hold-out testing approach. The dataset was divided into separate training and testing subsets to ensure that the models were evaluated on unseen data. This approach helps assess the model's ability to generalize beyond the training dataset and reduces the risk of overfitting. The classification performance was assessed using accuracy, precision, recall, and F1-score, providing a comprehensive evaluation of model effectiveness.

GitHub Repository Link for the codes developed: <https://github.com/pisengupta/04-Fake News Detection and Evaluation with Confusion Matrix Summer 2026>

5. Data Analysis and Results

5.1 Descriptive Analysis

The dataset consists of two classes of news articles: **fake news (class = 1)** and **true news (class = 0)**. After merging and preprocessing, the dataset contained a balanced distribution of both classes, which is suitable for classification tasks.

A basic exploration of the dataset showed that the text data varies in length and complexity, with news articles covering a wide range of topics. After cleaning, only the textual content and class label were retained for analysis.

5.2 Dataset Summary

Dataset Component	Description
Total Records	~44,000+ articles
Features Used	Text
Target Variable	Class (0=True, 1=Fake)
Missing Values Removed	633 rows
Final Dataset Shape	Clean and balanced

5.3 Class Distribution

The dataset contains two nearly balanced classes:

Class	Label	Description
0	True News	Genuine news articles
1	Fake News	Misleading or fabricated news

This balance ensures that the model is not biased toward any one class.

5.4 Model Performance Comparison

The performance of the Machine Learning models is summarised below:

Model	Accuracy
Logistic Regression	0.99 (approx.)
Decision Tree Classifier	0.9954

Interpretation:

- The Logistic Regression model performed extremely well due to its efficiency in handling high-dimensional sparse text data.
- The Decision Tree Classifier model achieved slightly higher accuracy but may be more prone to overfitting compared to that of the Logistic Regression model.

5.5 Detailed Evaluation (Logistic Regression)

The Logistic Regression model was evaluated using precision, recall, and F1-score.

Class	Precision	Recall	F1-Score	Support
0 (True)	0.99	0.99	0.99	5333
1 (Fake)	0.99	0.99	0.99	5738

Overall Metrics:

- Accuracy: 99%
- Macro Average F1-Score: 0.99
- Weighted Average F1-Score: 0.99

Interpretation:

The model demonstrates strong and balanced performances across both the classes, indicating that it is equally effective in identifying fake and true news articles.

5.6 Decision Tree Model Performance

The Decision Tree Model achieved:

- Accuracy: 0.9954

This indicates that the model correctly classified approximately 99.54% of test samples. However, Decision Trees are generally more sensitive to small variations in data and may overfit compared to linear models like Logistic Regression.

5.7 Key Observations

- TF-IDF representation was highly effective in transforming textual data into meaningful numerical features.
- Both models performed exceptionally well, with accuracies above 99%.
- Logistic Regression provided stable and interpretable results.
- Decision Tree achieved slightly higher accuracy but may not generalize as consistently as Logistic Regression.
- The preprocessing steps significantly contributed to model performance by reducing noise in the dataset.

5.8 Conclusion from Analysis

The results clearly demonstrate that Machine Learning techniques can effectively distinguish between fake and genuine news articles. The combination of TF-IDF and supervised classification models resulted in high predictive accuracy, making the system suitable for real-world fake news detection tasks.

6. Conclusion

This project successfully demonstrates the application of Machine Learning techniques for the classification of news articles into fake and genuine categories. Starting from raw textual data, the complete pipeline—from preprocessing to model evaluation—highlights how structured learning systems can be built to address real-world problems such as misinformation detection.

The results obtained are highly encouraging. The Logistic Regression model achieved an accuracy of approximately **99%**, with consistently strong precision, recall, and F1-scores across both classes. The Decision Tree Classifier also performed competitively, reaching an accuracy of about **99.54%**. These results indicate that the TF-IDF representation effectively captures meaningful patterns in textual data, allowing even relatively simple models to achieve strong predictive performance.

An important observation from this study is that Logistic Regression provides more stable and interpretable performance in high-dimensional sparse text settings, while Decision Trees, although slightly more accurate in this case, may be more sensitive to variations in the dataset. This reinforces the idea that model selection in text classification should consider not only accuracy but also generalization behavior.

Overall, the project highlights that a well-structured machine learning pipeline—combining data cleaning, TF-IDF feature extraction, and supervised learning—can achieve highly reliable results in fake news detection without requiring overly complex models.

Future Scope

While the current system performs effectively on the given dataset, there is significant scope for enhancement. More advanced language models such as BERT or transformer-based architectures could be explored to capture deeper contextual meaning beyond word frequency. Additionally, integrating ensemble methods may further improve robustness.

In practical applications, this model could be extended into a real-time fake news detection tool or integrated into social media platforms to assist in content verification. Expanding the dataset to include more diverse and multilingual sources would also improve generalization and real-world applicability.

In essence, this project not only achieves its immediate objective but also provides a strong foundation for more advanced research in automated misinformation detection using machine learning

7. APPENDICES

7.1 APPENDIX A: References

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- 7.1.9 Fake and Real News Datasets. Kaggle. Available at: <https://www.kaggle.com>

7.2 APPENDIX B: GitHub Repository Link

https://github.com/pisengupta/04-Fake_News_Detection_and_Evaluation_with_Confusion_Matrix_Summer_2026